

PROJECT REPORT GROUP 10

DESIGN AND ANALYSIS OF (3 BIT) R-2R LADDER [DAC]



GROUP MEMBERS

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ABSTRACT:

Data Converters

The electronic circuits, which can be operated with analog signals are called as analog circuits. Similarly, the electronic circuits, which can be operated with digital signals are called as digital circuits. A data converter is an electronic circuit that converts data of one form to another.

There are two types of data converters –

- Analog to Digital Converter
- Digital to Analog Converter

- **ANALOG TO DIGITAL CONVERTER:**

The circuit that converts the analog signal into digital signal is called as Analog to Digital Converter.

- **DIGITAL TO ANALOG CONVERTER:**

The circuit that converts the digital signal into an analog signal is called as Digital to Analog Converter

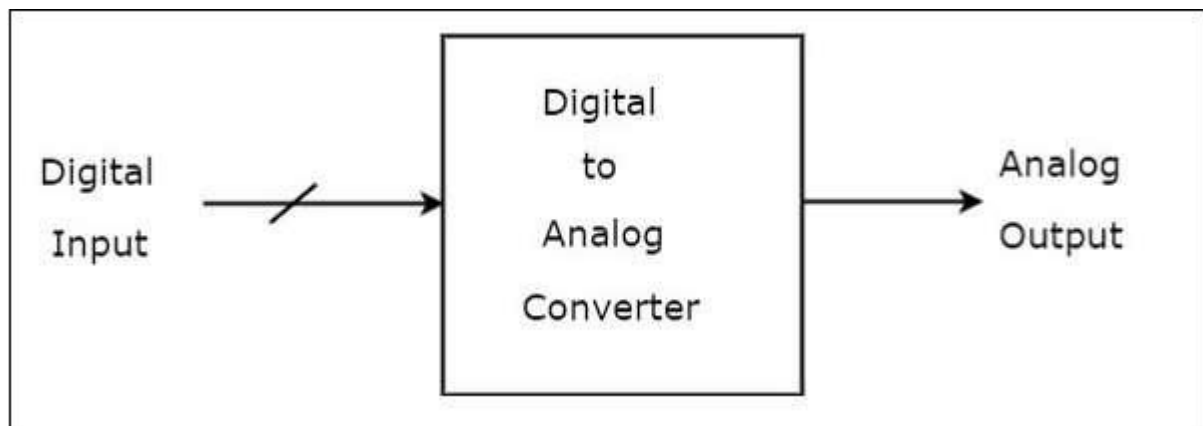
INTRODUCTION

What is a DAC?

The **DIGITAL TO ANALOG CONVERTER (DAC)** converts a given input signal into analog output signal. The digital signal is represented with a binary code which is a combination of bits 0 and 1.

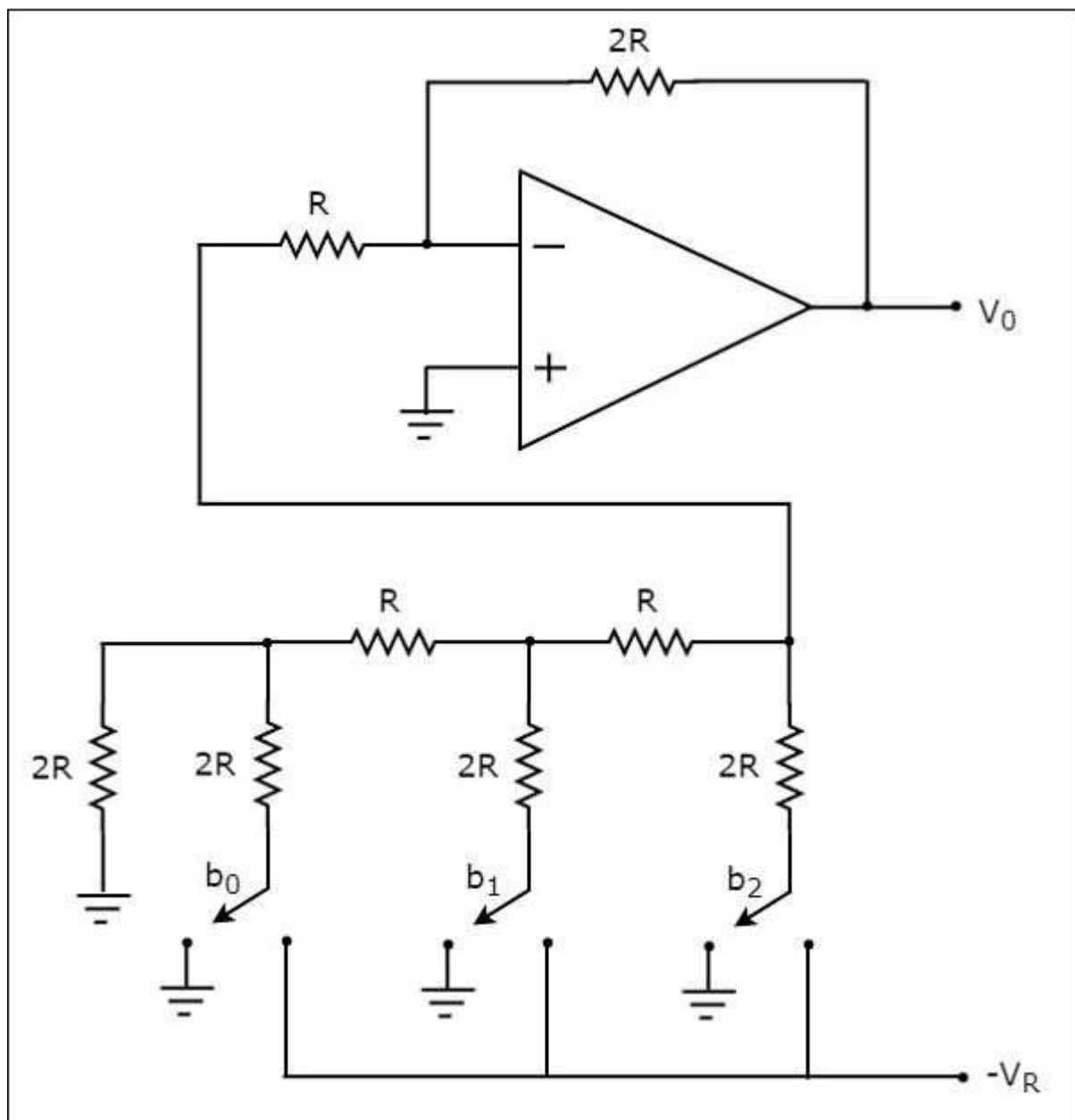
A digital to analog converter consists of many number of binary inputs and a single output .The number of binary outputs in a DAC will be a power of 2

BLOCK DIAGRAM:



3 BIT R-2R LADDER

A 3-bit R-2R ladder is a DAC that converts a 3-bit digital input into an analog output. Its design relies on resistors arranged in a repeating pattern of R and 2R values. This simplifies the circuit and makes it easy to scale for higher bit resolutions.



THEORY

The input lines b_0, b_1, b_2 are the three digital bits with b_2 being the most significant bit (MSB) and b_0 being the least significant bit (LSB). R and $2R$ resistors are arranged in a ladder structure. The pattern repeats in such a way that each subsequent bit contributes to the analog output. An op-amp is used at the output to buffer the signal and improve accuracy. V_R Provides the voltage level that the digital bits scale down to create the analog output.

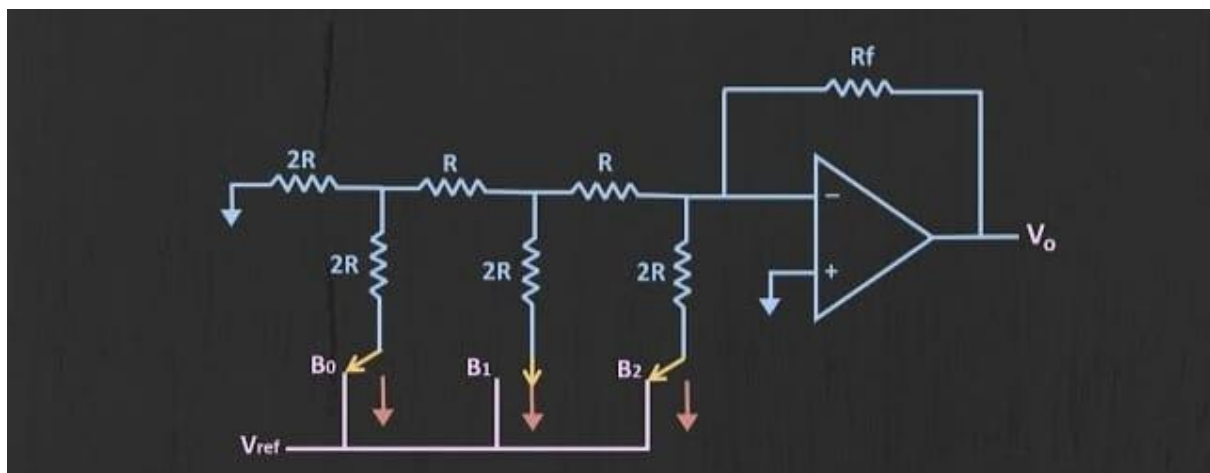
WORKING PRINCIPLE

- Each bit in the input determines whether a corresponding fraction of the reference voltage is included in the output.
- The resistive ladder ensures that higher bits have a larger influence on the output than lower bits.
- The analog output is the sum of the voltages contributed by each bit.
- This design ensures accurate digital-to-analog conversion using only two resistor values and can be easily expanded for more bits.

COMPONENTS REQUIRED FOR THE
EXPERIMENT :

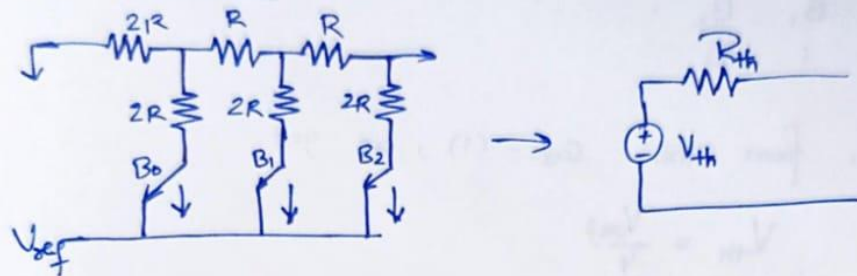
S.No	Name of the component	Range	Quantity
1	Resistors	2.2k 4.7k	3 5
2	Switches	-	3
3	RPS	-	1
4	IC741 (Op-Amp)	-	1
5	Multimeter	-	1
6	Connecting Wires	-	10

CIRCUIT DIAGRAM



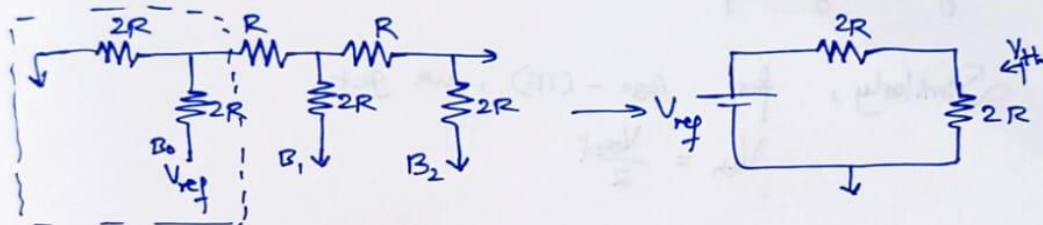
THEORETICAL CALCULATION:

* Theory Calculation :-

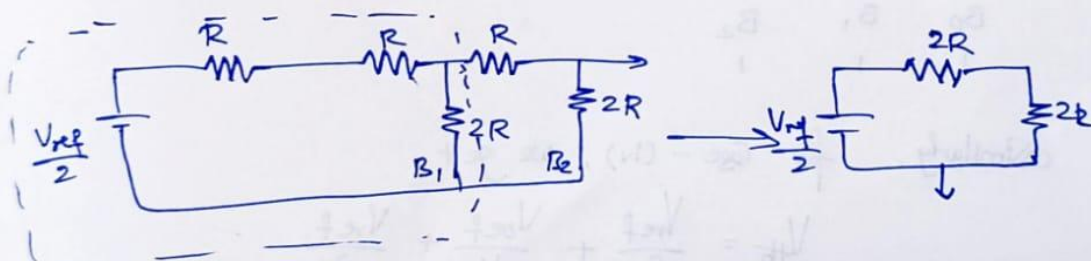


Case - (i) :-

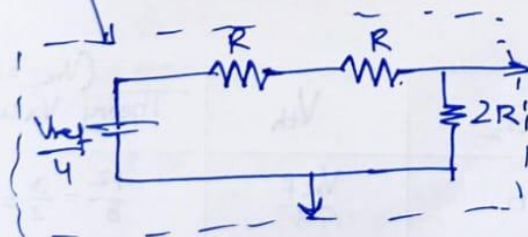
$B_0 \quad B_1 \quad B_2$
 $\quad 1 \quad 0 \quad 0$



$$V_{th} = \frac{2R \times V_{ref}}{2R + 2R} = \frac{V_{ref}}{2}, \quad R_{th} = 2R \parallel 2R = R.$$



$$V_{th} = \frac{2R \times (V_{ref}/2)}{2R + 2R} = \frac{V_{ref}}{4}, \quad R_{th} = 2R \parallel 2R = R.$$



$$V_{th} = \frac{2R \times (V_{ref}/4)}{2R + 2R} = \frac{V_{ref}}{8}, \quad R_{th} = 2R \parallel 2R = R.$$

Case - (i) :-

$$\begin{matrix} B_0 & B_1 & B_2 \\ 0 & 1 & 0 \end{matrix}$$

Similarly, from the case - (i), we get

$$V_{th} = \frac{V_{ref}}{4}$$

Case - (ii) :-

$$\begin{matrix} B_0 & B_1 & B_2 \\ 0 & 0 & 1 \end{matrix}$$

Similarly, for case - (ii), we get

$$V_{th} = \frac{V_{ref}}{2}$$

Case - (iv) :-

$$\begin{matrix} B_0 & B_1 & B_2 \\ 1 & 1 & 1 \end{matrix}$$

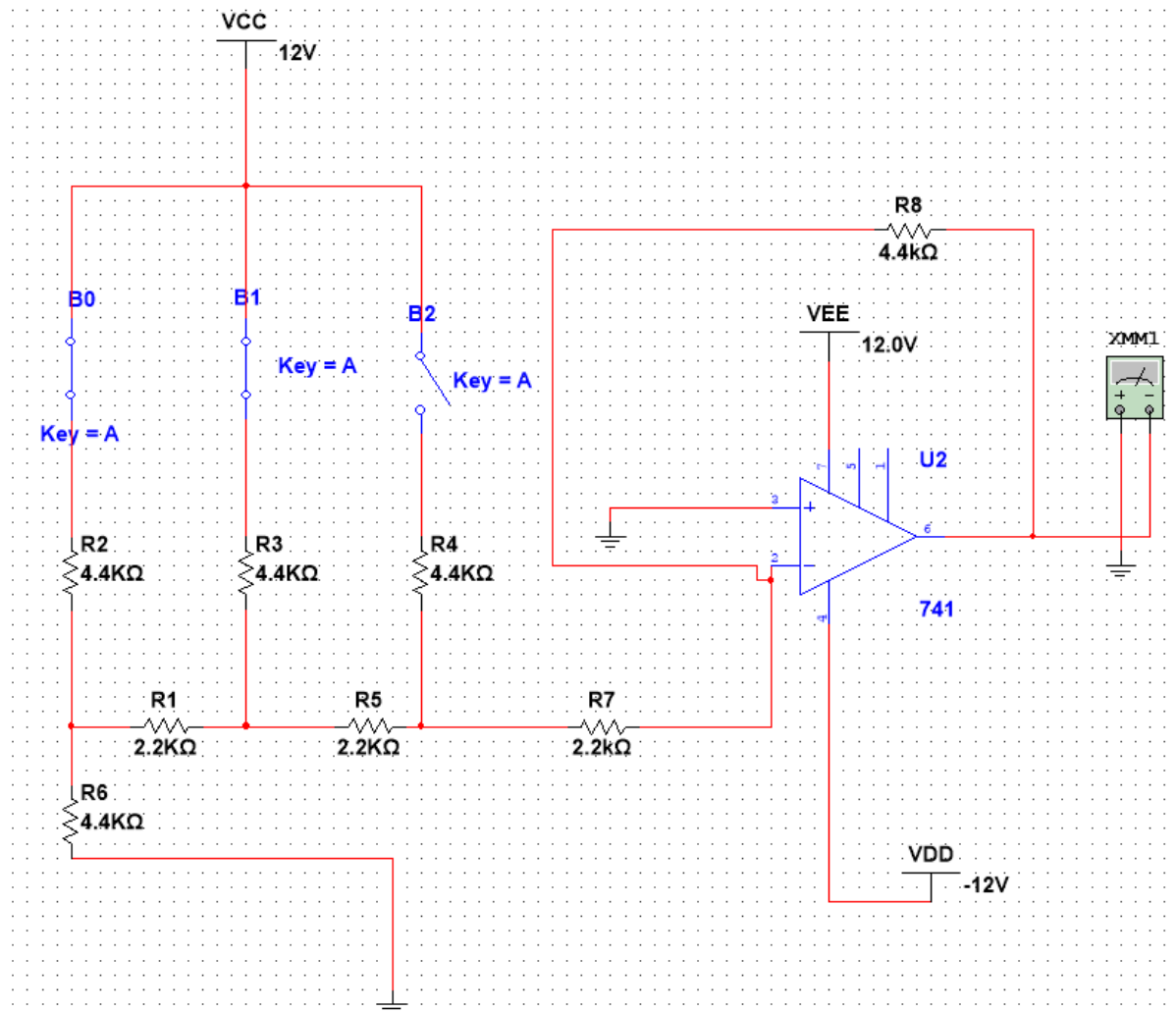
Similarly, for case - (iv), we get

$$\begin{aligned} V_{th} &= \frac{V_{ref}}{8} + \frac{V_{ref}}{4} + \frac{V_{ref}}{2} \\ &= \frac{7V_{ref}}{8} \end{aligned}$$

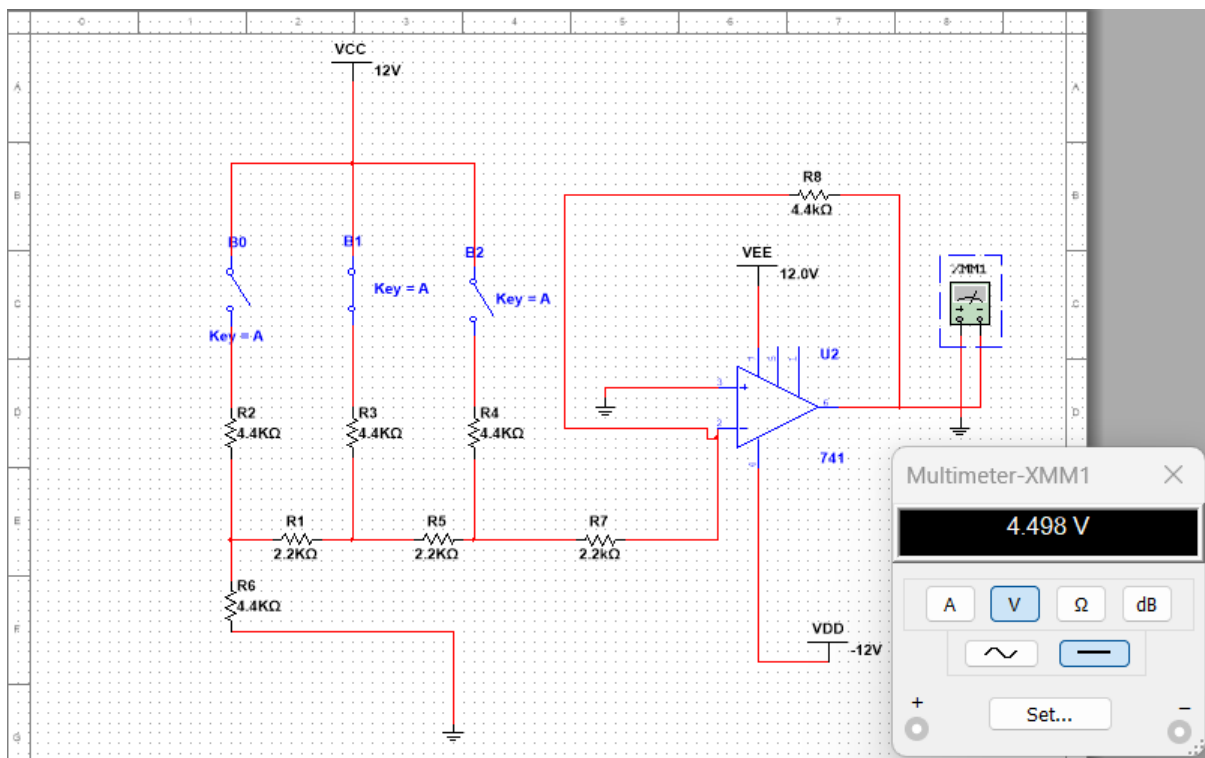
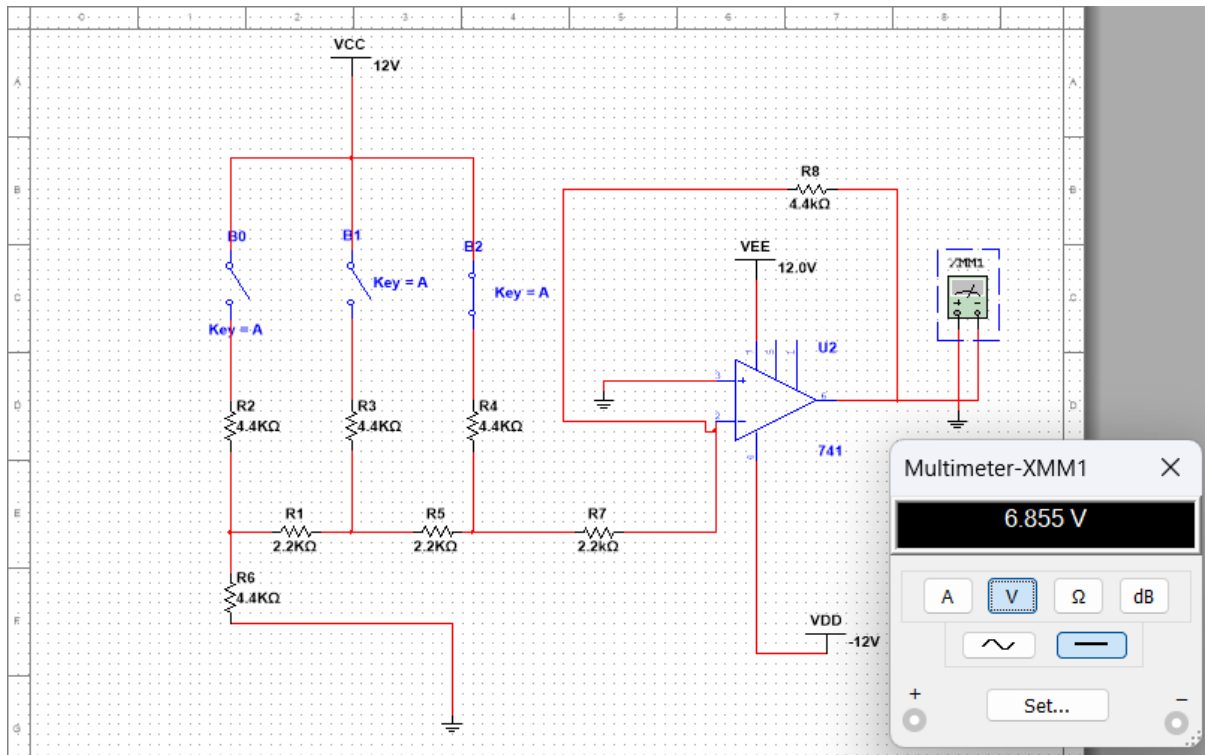
From above Cases,

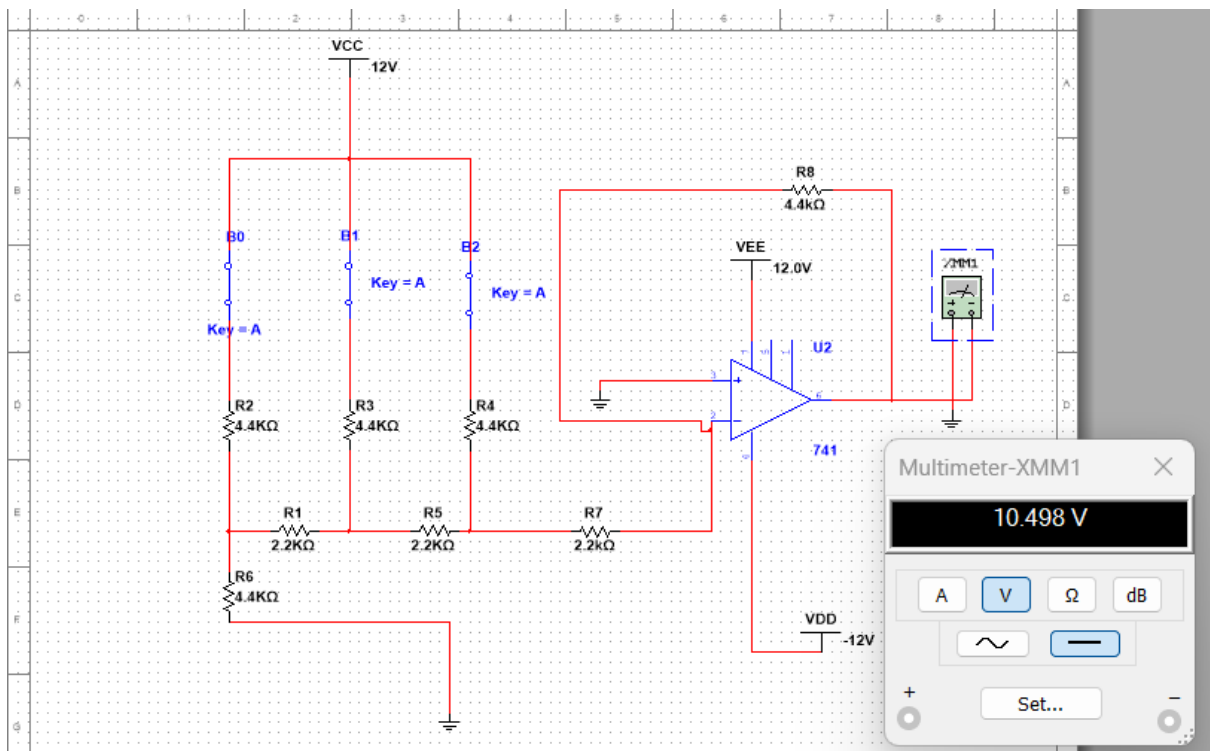
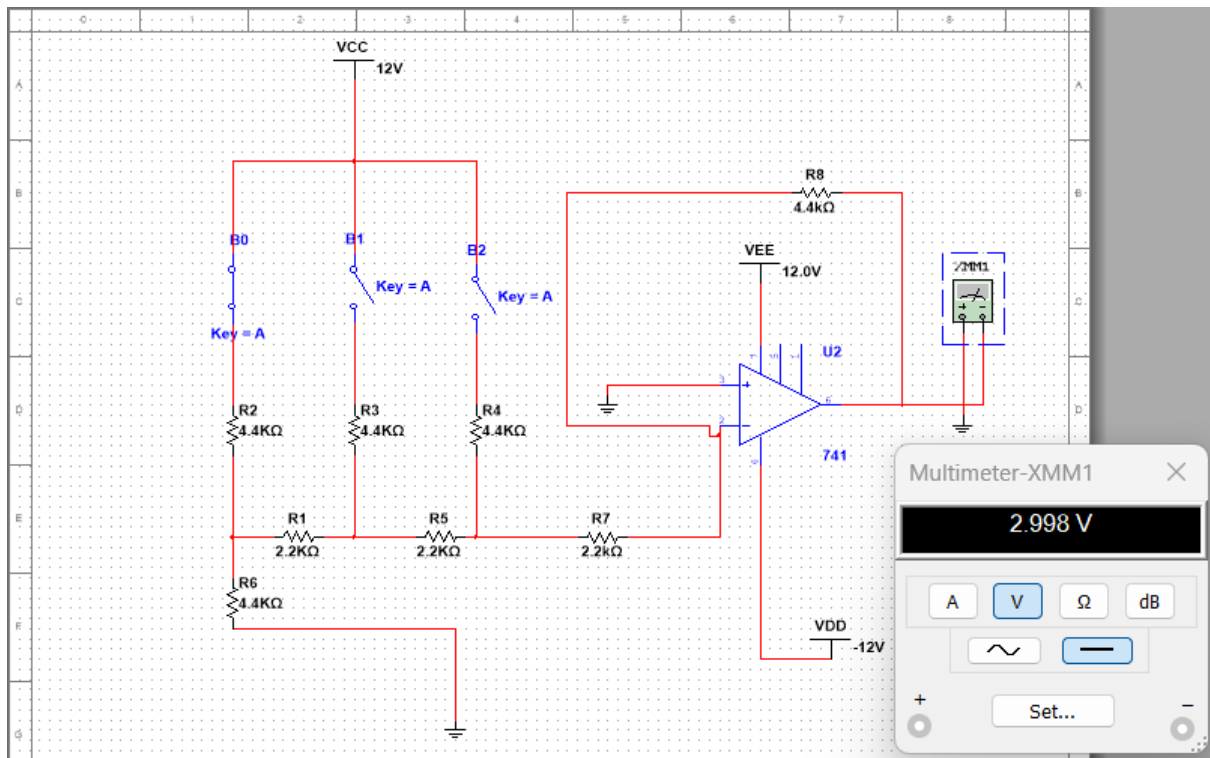
	B_0	B_1	B_2	V_{th}	$(V_{ref} = 12V)$ Theory Values.
i)	1	0	0	$\frac{V_{ref}}{8}$	$\frac{12}{8} = \frac{3}{2} = 1.5V$
ii)	0	1	0	$\frac{V_{ref}}{4}$	$\frac{12}{4} = 3V$
iii)	0	0	1	$\frac{V_{ref}}{2}$	$\frac{12}{2} = 6V$
iv)	1	1	1	$\frac{7V_{ref}}{8}$	$\frac{7 \times 12}{8} = 10.5V$

SIMULATION IN MULTISIM

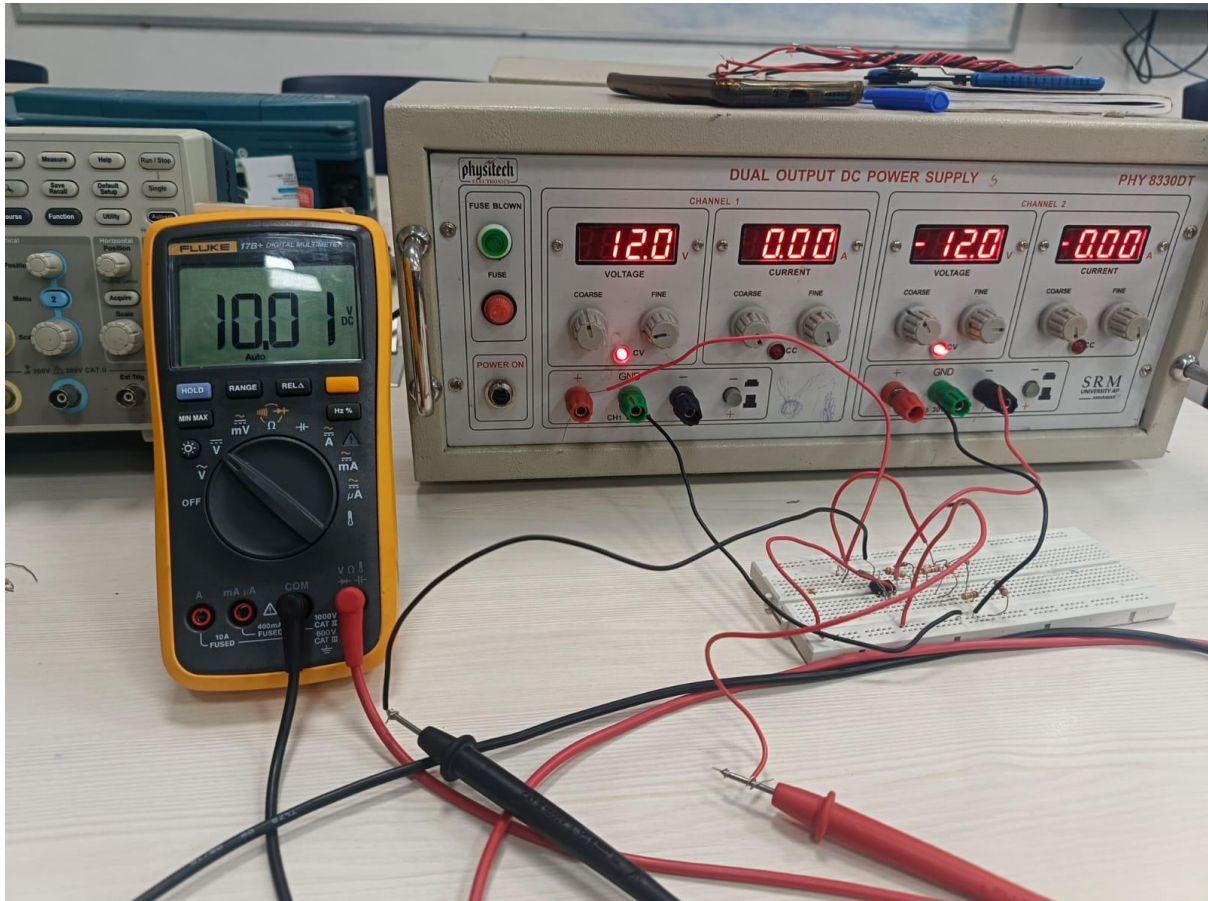


MULTISIM RESULT:

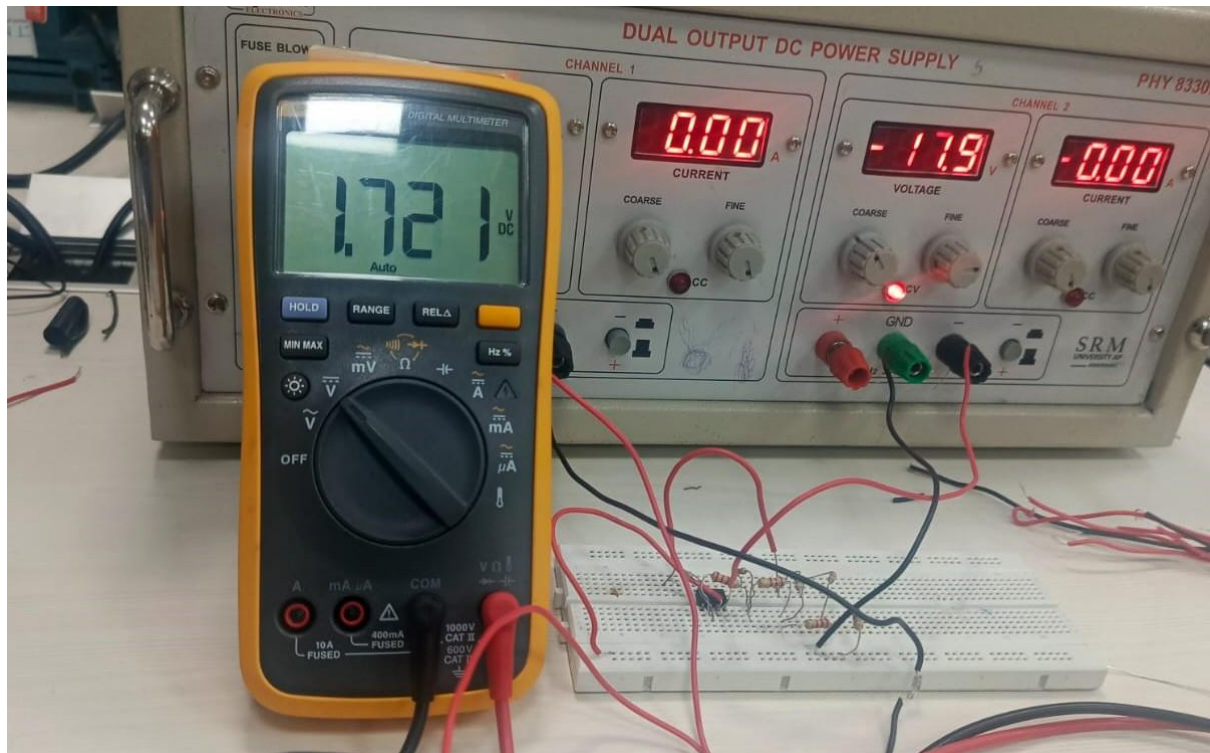




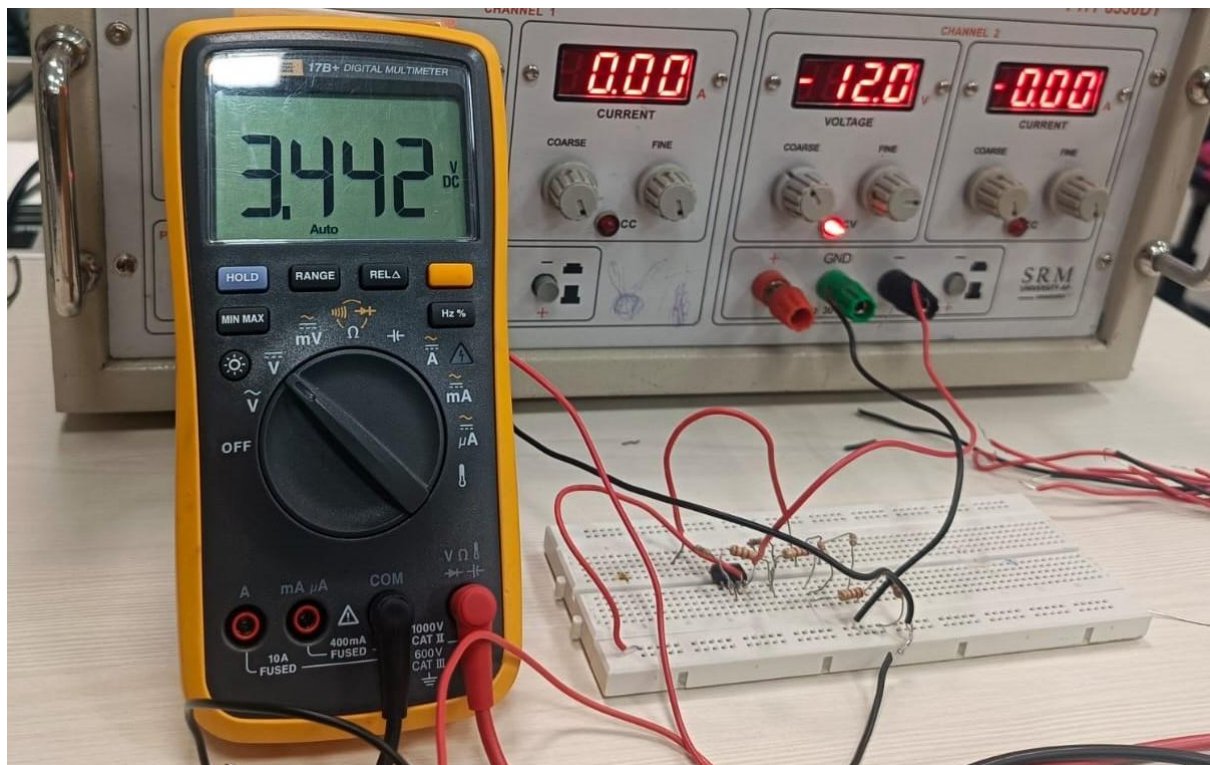
PRACTICAL EXPERIMENT:



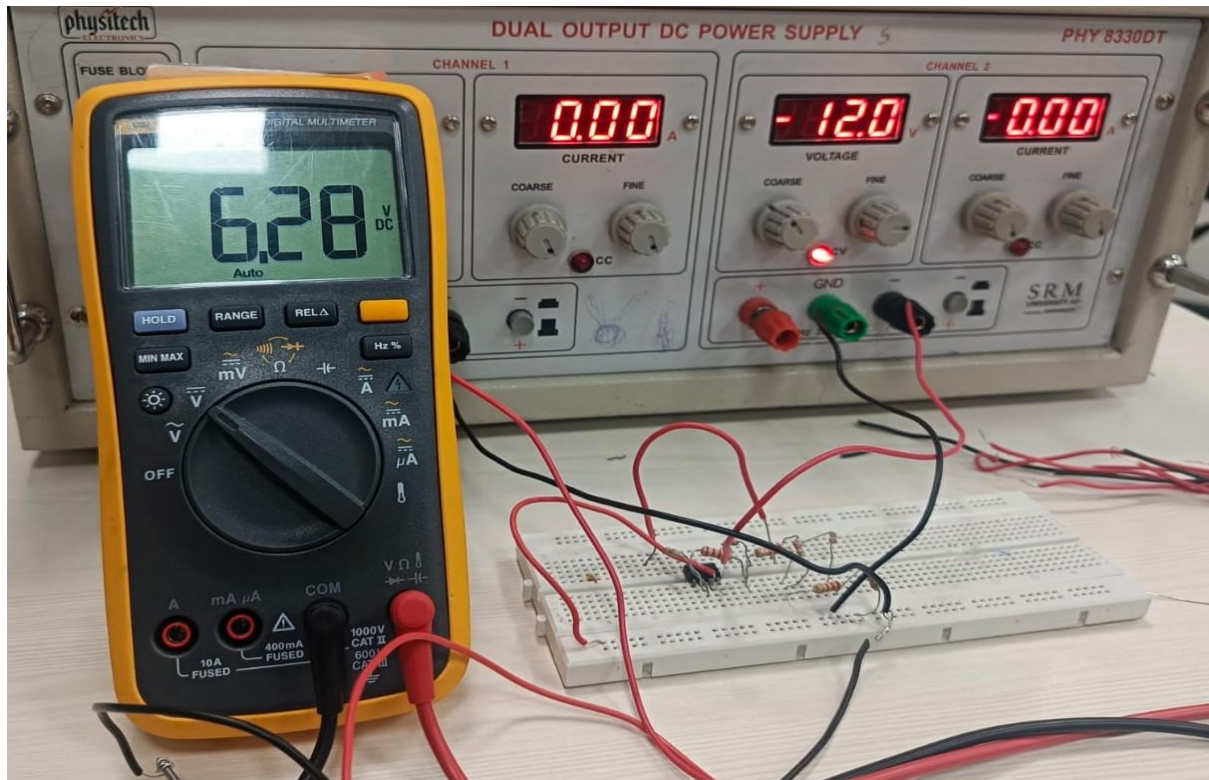
CASE-1: $B_0 = 1$ $B_1 = 0$ $B_2 = 0$



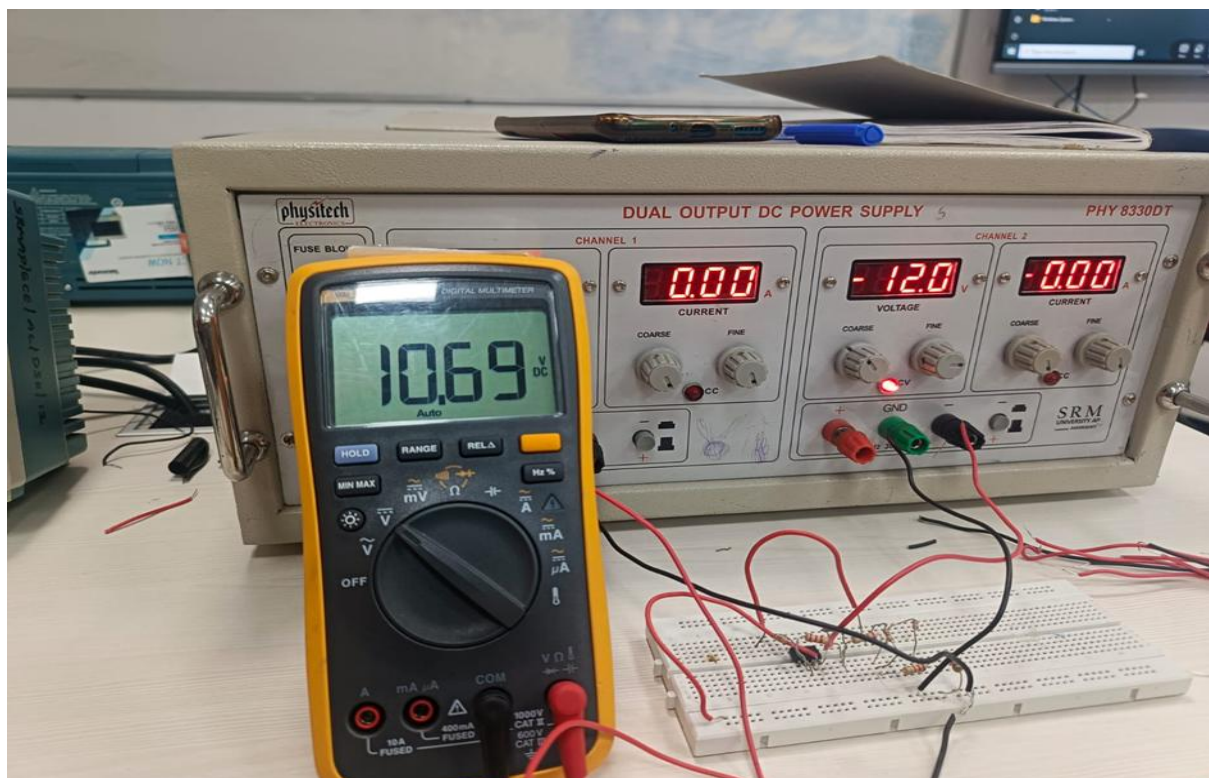
CASE-2: $B_0 = 0$ $B_1 = 1$ $B_2 = 0$



CASE-3: $B_0 = 0$ $B_1 = 0$ $B_2 = 1$



CASE-4: $B_0 = 1$ $B_1 = 1$ $B_2 = 1$



CONCLUSION :

The 3-bit R-2R ladder DAC provides an efficient way to convert 3-bit digital signals to analog but is better suited for low-resolution, low-cost applications where simplicity and power efficiency are more important than high precision.

We have studied the voltage response of 3bit R-2R Ladder design using operational-amplifier and verified the theoretical voltage values using Multisim. We used Thevenin's Theorem to calculate the theoretical values.

We have verified Four Cases:

Case-1 $B_0=1, B_1=0, B_2=0$

Case-2 $B_0=0, B_1=1, B_2=0$

Case-3 $B_0=0, B_1=0, B_2=1$

Case-4 $B_0=1, B_1=1, B_2=1$